Kevin Lam

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EDUCATION

University of Oxford

PhD in Statistical Machine Learning — Oct 2024 – Apr 2028 (expected)

- Research Areas: Generative Modelling, Sampling, Optimisation, Numerical Methods
- Supervisors: Yee Whye Teh & George Deligiannidis
- Scholarships: Google DeepMind PhD Scholarship

University of New South Wales

Master of Mathematics — Jun 2020 – Dec 2023

- Research Areas: Proper Scoring Rules, Invertible Neural Networks, Statistical Computing
- Focus Areas: Mixed-integer, Conic and Linear Programs, Stochastic Processes, Multivariate Analysis

Master of Information Technology — Jul 2017 – Jan 2020

- Research Areas: Reinforcement Learning, Imitation Learning, Generative Adversarial Networks
- Focus Areas: Design & Analysis of Algorithms, Machine Learning

Bachelor of Commerce/Science — Feb 2011 – Dec 2015

PUBLICATIONS & PREPRINTS

- *Metropolis-Adjusted Diffusion Models*, **K. H. Lam***, T. Farghly*, C. Williams*, J. Yang, Y. W. Teh, A. Doucet, Conference on Neural Information Processing Systems 2026, NeurIPS 2026
- *Rao-Blackwellised Reparameterisation Gradients*, **K. H. Lam**, T. D. Bui, G. Deligiannidis, Y. W. Teh, Conference on Neural Information Processing Systems 2025, NeurIPS 2025
- *Regularized Black-box Optimization Algorithms for Least-squares Problems*, Y. Liu, **K. H. Lam**, L. Roberts, IMA Journal of Numerical Analysis, IMAJNA 2025
- *LegendreTron: Uprising Proper Multiclass Loss Learning*, **K. H. Lam**, C. Walder, S. Penev, R. Nock, International Conference on Machine Learning 2023, ICML 2023

RESEARCH EXPERIENCE

Microsoft – *Research Intern* — Apr 2026 – Jul 2026

- Research on Image Editing with Diffusion and Flow models

Australian National University – *Research Intern | Research Assistant* — Nov 2022 – Oct 2024

- Research on Variational Inference for Discrete Neural Networks

University of Technology Sydney – *Research Assistant* — Nov 2023 – Apr 2024

- Research on Structure Learning for Bayesian Networks and LLMs

Australian National University – *Research Intern* — Nov 2023 – Jan 2024

- Research on Control Theory and Differential Geometry

Australian National University – *Research Intern* — Nov 2021 – Jan 2022

- Research on Regularised Black-box Optimisation algorithms

Department of Defence (DSTG) – *Research Cadet* — May 2021 – Nov 2021

- Research on Adversarial Machine Learning

University of Sydney – *Research Intern | Research Engineer* — Dec 2020 – May 2021

- Research on Manifold MCMC algorithms

SKILLS

- Programming & Tooling – Python, SQL, R, Matlab, C, C++
- Python Libraries – PyTorch, JAX, Tensorflow, XGBoost, LightGBM, scikit-learn, pandas, NumPy, SciPy
- Tooling – Linux, Git, Kubernetes, Docker, Docker Compose
- Computing Services – Databricks, Spark, Azure, AWS, GCP, HPC, Slurm

TEACHING

SC5 Advanced Simulation Methods – *Teaching Assistant* Jan 2026 – Mar 2026
SB2.2 Statistical Machine Learning – *Teaching Assistant* Jan 2025 – Mar 2025
COMP9444 Neural Nets & Deep Learning – *Consultation Tutor* Sep 2020 – Dec 2020

INDUSTRY EXPERIENCE

QBE – *Machine Learning Engineer* Jan 2024 – Sep 2024

- Automated **Docker** workflows on **JFrog Artifactory** using **Azure pipelines** as CI/CD
- Deployed a database backup service using **Docker Compose** and **CRON**

Hivery – *Machine Learning Engineer* Aug 2022 – Jan 2024

- Led the adoption of **MLOps** practices by using **Azure Databricks** and **MLflow** to save and track models
- Automated building and storage of model container images by using **Docker** and **Gitlab pipelines** as CI/CD
- Implemented and deployed demand transfer models for retail assortment

Woolworths (WooliesX) – *Senior Data Scientist* Jul 2018 – Oct 2019

- Implemented **GLM** and **GBM** models for loyalty programs including **Personalisation** and **Customer Lifetime Value**
- Illustrated effectiveness of ML-driven marketing through a **randomised controlled trial** and **A/B testing**

Quantium – *Data Analyst* Nov 2017 – May 2018

- Identified home delivery expansion opportunities for a national retailer by using analytics generated from **PySpark**

Commonwealth Bank – *Quantitative Analyst* Jan 2017 – Oct 2017

- Developed **IRB models** for regulatory capital by modelling **probabilities of default** against risk factors with **GLMs**

AMP – *Financial Risk Analyst* Jan 2016 – Dec 2016

PROFESSIONAL ACCREDITATIONS

Global Association of Risk Professionals – *Financial Risk Manager* Sep 2021 – Present
Statistical Society of Australia – *Graduate Statistician* Nov 2020 – Dec 2023
CFA Institute – *Chartered Financial Analyst* Sep 2020 – June 2021